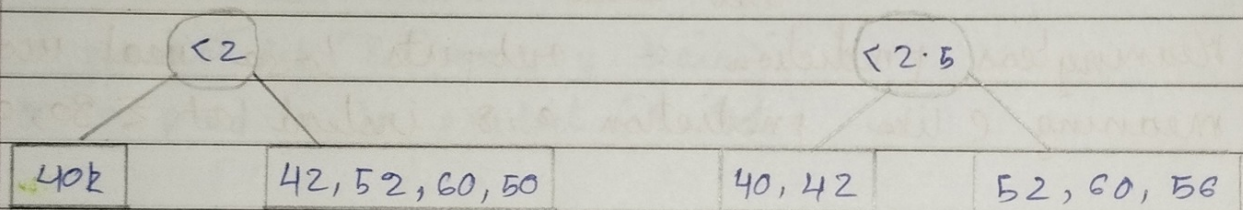


Decision Tree Regression

Dataset (Regression)

Exp	Grp	Salary
2	Yes	40
2.5	Yes	42
3	No	52
4	No	60
4.5	Yes	56

Let's say we have 2 ~~to~~ split and we need to choose the one with **highest variance reduction value**



- ⇒ Here, we have 2 split now our goal is to choose the one which is best, but how?
- ⇒ In classification we do this with the help of Gini Index, Entropy & Information Gain
- ⇒ In regression we do this using **Variance Reduction**

Step 1 :- Calculate Variance for each node

Formula :-

$$\Rightarrow \text{Variance} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2 \quad \hat{y} = \text{Average o/p}$$

a) Calculate Variance for root node

a) calculate variance for root node

Now, how many outputs root node (2) has?

⇒ We have 5 outputs :- 40, 42, 52, 60, and 56

$\hat{y}$ = Average of output

$$= \frac{40 + 42 + 52 + 60 + 56}{5} = \frac{250}{5}$$

$$\hat{y} = 50$$

$$\text{Variance (Root)} = \frac{1}{5} [(40 - 50)^2 + (42 - 50)^2 + (52 - 50)^2 + (60 - 50)^2 + (56 - 50)^2]$$

$$= \frac{1}{5} [100 + 64 + 4 + 100 + 36]$$

= 60.8 for root (2) and this is the same for (2.5) root

b) calculate Variance for child nodes

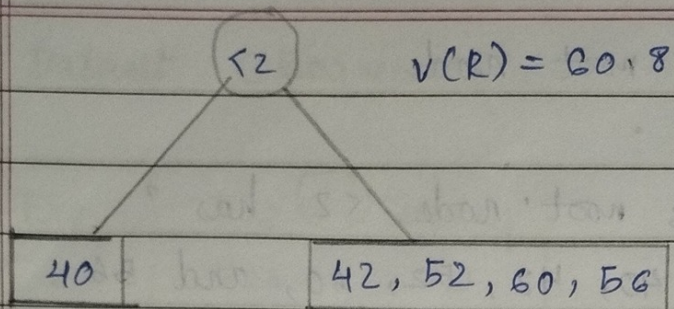
$$\text{Variance (Child 1)} = \frac{1}{1} [(40 - 50)^2] = 100$$

$$\text{Variance (Child 2)} = \frac{1}{4} [(42 - 50)^2 + (52 - 50)^2 + (60 - 50)^2 + (56 - 50)^2]$$

$$= \frac{1}{4} [64 + 4 + 100 + 36]$$

$$= 51$$

[40, 42, 52, 60, 56]



$$V(R) = 60.8$$

Variance Reduction
of this node is 0

$$V(C1) = 100$$

$$V(C2) = 51$$

Step 2 :- Variance Reduction

Formula :-

$$= V(R) - \sum W_i V(C) \quad \text{or} \quad \text{Var}(\text{Root}) - \sum W_i \text{Var}(\text{child})$$

$$= 60.8 - \left[\frac{1}{5} (100) + \frac{4}{5} (51) \right]$$

What is W_i ?

$$= 60.8 - 20 - 40.8$$

⇒ After splitting how many
nodes child1 has compared
to root node

$$= 0$$

— Now let's calculate variance of child for second node

$$V(C1) = \frac{1}{2} [(40-50)^2 + (42-50)^2]$$

$$= \frac{1}{2} [100 + 64] = \frac{164}{2} = 82$$

$$V(C2) = \frac{1}{3} [(52-50)^2 + (60-50)^2 + (56-50)^2]$$

$$= \frac{1}{3} [4 + 100 + 36] = 46.66$$

Variance Reduction for Node 2

[40, 42, 52, 60, 56]

≤ 2.5 $V(R) = 60.8$

40, 42

52, 60, 56

$V(C1) = 82$

$V(C2) = 46.66$

Variance Reduction Formula :-

$$VR = V(R) - \sum W_i V(C)$$

$$= 60.8 - \left[\frac{2}{5} (82) + \frac{3}{5} (46.66) \right]$$

$$= \cancel{0.004} \quad 0.004$$

Node1 = 0

Node2 = 0.004

Therefore, we choose the split with higher variance reduction :- **node2 > node1**, we split with node2 for better results